3.5 Undetermined Coefficients

In each of Problems 1 through 14, find the general solution of the given differential equation.

3.5. 7. $y'' + 9y = t^2 e^{3t} + 6$ $\quad y'' + 9y = t^2 e^{3t} \to Y_2 \quad y'' + 9y = 6 \to Y_1$

hom. DE: $y'' + 9y = 0$ $\quad 2^{nd}$ order lin, const coef.

char eqn: $r^2 + 9 = 0 \to r_{1,2} = \pm 3i \to y_1 = \cos 3t,\ y_2 = \sin 3t$

hom. soln: $y = c_1 \cos 3t + c_2 \sin 3t$.

(1) nonhom: $y'' + 9y = 6$ ($P(t)$ of deg 0) — method of undet coeffs →

$Y_1 = P(t)$ of deg $0 \cdot t^s$ (s is the number of times 0 is a root of char eqn)
$= C$. ($s=0$ here)

$9C = 6 \to C = 2/3$ $\quad y' = 0$

$Y_1'' + 9Y_1 = 6 \longrightarrow$ soln of 1st part

$+\ Y_2'' + 9Y_2 = t^2 e^{3t} \longrightarrow$ soln of 2nd part

$(Y_1 + Y_2)'' + 9(Y_1 + Y_2) = t^2 e^{3t} + 6 \longrightarrow y_{particular} = Y_1 + Y_2$

(2) nonhom: $y'' + 9y = t^2 e^{3t} = \underset{\deg 2}{P(t)} \cdot e^{3t} \cdot t^s \longrightarrow$ s is the number of times 3 is a soln of char eqn. ($s=0$, here.)

$Y_2 = (At^2 + Bt + C) \cdot e^{3t}$

$Y_2'' + 9Y_2 = t^2 e^{3t}$

$Y_2' = (2At + B)e^{3t} + 3(At^2 + Bt + C)e^{3t}$

$= e^{3t}(3At^2 + (2A+3B)t + B + 3C)$

$Y_2'' = 3e^{3t}(3At^2 + (2A+3B)t + B + 3C) + e^{3t}(6At + 2A + 3B)$

$t^2 e^{3t} = 3e^{3t}(3At^2 + (2A+3B)t + B + 3C) + e^{3t}(6At + 2A + 3B) + 9(At^2 + Bt + C) \cdot e^{3t}$

$t^2 = (9A + 9A)t^2 + (6A + 9B + 6A + 9B)t + 3B + 9C + 2A + 3B + 9C$

(① $= 1$, ② $= 0$, ③ $= 0$)

① $18A = 1 \rightarrow A = 1/18$

② $12A + 18B = 0$

$1/9 = -3B \rightarrow B = -1/27$

③ $-\frac{2}{9} + 18C + \frac{1}{9} = 0$

$18C = \frac{1}{9} \quad C = \frac{1}{162}$

$y_1 = \frac{2}{3} \qquad y_2 = \left(\frac{t^2}{18} - \frac{t}{27} + \frac{1}{162}\right)e^{3t}$

$$y_{part} = \frac{2}{3} + \left(\frac{t^2}{6} - \frac{t}{27} + \frac{1}{162}\right)e^{3t}$$

gen. soln: $y = y_{hom} + y_{par}$

$$= c_1 \cos 3t + c_2 \sin 3t + \frac{2}{3} + \left(\frac{t^2}{6} - \frac{t}{27} + \frac{1}{162}\right)e^{3t}$$

In each of Problems 15 through 20, find the solution of the given initial value problem.

3.5 16. $y'' + 4y = t^2 + 3e^t, \quad y(0) = 0, \quad y'(0) = 2$

hom DE: $y'' + 4y = 0$

char. eqn: $r^2 + 4 = 0 \rightarrow r_{1,2} = \pm 2i$ → $y_1 = \cos 2t$, $y_2 = \sin 2t$

$$y_{hom} = c_1 \cos 2t + c_2 \sin 2t$$

nonhom DE: $y'' + 4y = t^2 + 3e^t$

but $s=0$ since $r_{1,2} \neq 0$. but $q=0$ since $r_{1,2} \neq 1$

$$Y = y_{par} = (At^2 + Bt + C)t^s + De^t t^q$$

$$Y' = 2At + B + De^t$$

$$Y'' = 2A + De^t$$

$$Y'' + 4Y = 2A + De^t + 4(At^2 + Bt + C) + 4De^t = t^2 + 3e^t$$

$$(4A)t^2 + (4B)t + (2A + 4C) + (5D)e^t = (1)t^2 + (3)e^t$$

$$(0)t + (0)$$

$4A = 1 \rightarrow A = 1/4 \quad 4B = 0 \rightarrow B = 0 \quad \overset{1/2}{2A + 4C = 0} \rightarrow C = -1/8$

$5D = 3 \rightarrow D = 3/5$

$$y_{part} = Y = \frac{t^2}{4} - \frac{1}{8} + \frac{3}{5}e^t$$

gen. soln $= y = y_{hom} + y_{part}$

$$= c_1 \cos 2t + c_2 \sin 2t + \frac{t^2}{4} - \frac{1}{8} + \frac{3}{5}e^t$$

$$y' = -2c_1 \sin 2t + 2c_2 \cos 2t + \frac{t}{2} + \frac{3}{5}e^t$$

IVs: $y(0) = 0 \rightarrow c_1 - \frac{1}{8} + \frac{3}{5} = 0 \rightarrow c_1 = -\frac{19}{40}$

$y'(0) = 2 \rightarrow 2c_2 + \frac{3}{5} = 2 \rightarrow c_2 = -\frac{7}{10}$

$$y = -\frac{19}{40}\cos 2t - \frac{7}{10}\sin 2t + \frac{t^2}{4} - \frac{1}{8} + \frac{3}{5}e^t$$

Determine a suitable form for $Y(t)$ if the method of undetermined coefficients is to be used.

3.5. 27. $y'' + 3y' + 2y = e^t(t^2 + 1)\sin 2t + 3e^{-t}\cos t + 4e^t$

is $1+2i$ a soln to char eqn? | is $-1+i$ a soln of char eqn? | is 1 a soln of char eqn?

} questions we need to ask to determine if there should be a t^s factor. (Check the table 3.5-1 on p. 182)

char eqn $r^2 + 3r + 2 = 0 \quad (r+2)(r+1) = 0 \quad r_1 = -2, r_2 = -1.$

$y_1 = e^{-2t} \quad y_2 = e^{-t}$

$$Y_{part} = (At^2 + Bt + C)e^t \sin 2t + (Dt^2 + Et + F)e^t \cos 2t + G e^{-t}\cos t + H e^{-t}\sin t + J e^t$$

$y = v_1 y_1 + v_2 y_2$, where v_1 & v_2 are fncs of t.

$y' = v_1' y_1 + v_1 y_1' + v_2' y_2 + v_2 y_2'$ → 1st constraint v_1' & v_2'

$y'' = v_1'' y_1 + v_1' y_1' + v_1' y_1' + v_1 y_1'' + v_2'' y_2 + v_2' y_2' + v_2' y_2' + v_2 y_2''$

$t^2 y'' - 2y = t^2(v_1' y_1' + v_1 y_1'' + v_2' y_2' + v_2 y_2'') - 2(v_1 y_1 + v_2 y_2)$

$= t^2(v_1' y_1' + v_2' y_2') = 3t^2 - 1$ → 2nd constraint

v_1' & v_2'

$t^2 y_1'' - 2y_1 = 0$

$t^2 v_1 y_1'' - 2 v_1 y_1 = 0$

4 $t^2 y_2'' - y_2 = 0$

$t^2 v_2 y_2'' - 2 v_2 y_2 = 0$

In each of Problems 13 through 20, verify that the given functions y_1 and y_2 satisfy the corresponding homogeneous equation; then find a particular solution of the given nonhomogeneous equation. In Problems 19 and 20, g is an arbitrary continuous function.

3.6 13. $t^2 y'' - 2y = 3t^2 - 1, \quad t > 0; \quad y_1(t) = t^2, \quad y_2(t) = t^{-1}$

$y = v_1 t^2 + v_2 t^{-1}$ & constraint: $v_1' y_1 + v_2' y_2 = 0$

$y' = v_1' t^2 + v_2' t^{-1}$ = 0 $\quad v_1' t^2 + v_2' t^{-1} = 0$

$+ v_1(2t) + v_2(-t^{-2})$

$y'' = 2v_1' t + 2v_1 - v_2' t^{-2} + 2v_2 t^{-3}$

$t^2 y'' - 2y = t^2(2v_1' t + 2v_1 - v_2' t^{-2} + 2v_2 t^{-3}) - 2(v_1 t^2 + v_2 t^{-1})$

$= 2v_1' t^3 - v_2' \quad = 3t^2 - 1$

① $v_1' t^2 + v_2' t^{-1} = 0 \xrightarrow{\cdot t > 0} v_1' t^3 + v_2' = 0 \quad v_2' = -t^2 + \frac{1}{3}$

② $2v_1' t^3 - v_2' = 3t^2 - 1$,

$+$

$3v_1' t^3 = 3t^2 - 1 \quad v_1' = \frac{1}{t} - \frac{1}{3t^3}$

$$v_1' = \frac{1}{t} - \frac{1}{3t^3} \qquad v_2' = -t^2 + \frac{1}{3}$$

$$v_1 = \ln t + \frac{1}{6t^2} + c_1 \qquad v_2 = -\frac{t^3}{3} + \frac{t}{3} + c_2$$

$$y(t) = v_1 y_1 + v_2 y_2 = t^2 \ln t + \frac{1}{6} + c_1 t^2 - \frac{t^2}{3} + \frac{1}{3} + c_2 t^{-1}$$

$$= t^2\left(\ln t - \frac{1}{3}\right) + \frac{1}{2} + c_1 t^2 + c_2 t^{-1}$$

particular — homo

In each of Problems 29 through 32, use the method outlined in Problem 28 to solve the given differential equation.

reduction of order

$y(t) = v(t) y_1(t)$

3.6. 29. $t^2y'' - 2ty' + 2y = 4t^2, \quad t > 0; \quad y_1(t) = t$

$$y = vy_1 = vt$$
$$y' = v't + v$$
$$y'' = v''t + v' + v'$$

$$t^2(v''t + 2v') - 2t(v't + v) + 2(vt) = 4t^2$$

$$v''t^3 = 4t^2$$

$$v'' = \frac{4}{t}$$

$$v' = 4\ln t + c_1$$

$$v = 4(t\ln t - t) + c_1 t + c_2$$

$$y(t) = vt = 4(t^2\ln t - t^2) + c_1 t^2 + c_2 t$$

Ex Var. Par. for 3rd order DE – how to choose constraints?

We have $y_{hom} = c_1 y_1 + c_2 y_2 + c_3 y_3$

Assume $Y(t) = u y_1 + v y_2 + w y_3$ (3 unknowns) where u, v, w are functions of x.

$$Y'(t) = u y_1' + v y_2' + w y_3' + \underbrace{u' y_1 + v' y_2 + w' y_3}_{=0 \text{ (1st condition)}}$$

$$Y''(t) = u y_1'' + v y_2'' + w y_3'' + \underbrace{u' y_1' + v' y_2' + w' y_3'}_{=0 \text{ (2nd cond.)}}$$

$$Y'''(t) = \underbrace{u y_1''' + v y_2''' + w y_3'''}_{\text{this part gives } 0 \text{ when we plug in}} + u' y_1'' + v' y_2'' + w' y_3''$$

Theorem 4.1.1 If the functions $p_1, p_2, \ldots, p_n$ and g are continuous on the open interval I, then there exists exactly one solution $y = \phi(x)$ of the differential equation (2) that also satisfies the initial conditions (3), where x_0 is any point in I. This solution exists throughout the interval I.

Ex & Uniq. n^{th} order linear DE

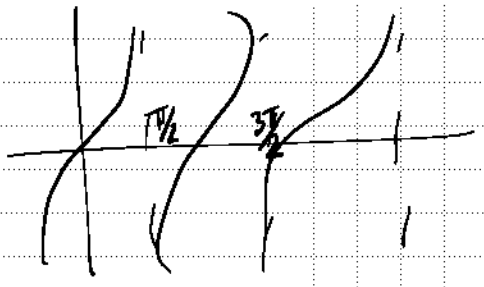
1. $y^{(n)} + p_1 y^{(n-1)} + p_2 y^{(n-2)} \cdots + p_n y = g$

In each of Problems 1 through 6, determine intervals in which solutions are sure to exist.

4.1.5. $(x-1)y^{(4)} + (x+1)y'' + (\tan x)y = 0$

4th order, lin. ODE

$$y^{(4)} + \underbrace{\frac{x+1}{x-1}}_{\text{discts if } x=1} y'' + \underbrace{\frac{\tan x}{x-1}}_{x=1 \text{ \& } x=\frac{\pi}{2}+k\pi,\ k\in\mathbb{Z}} y = 0$$



$\left(-\frac{3\pi}{2}, -\frac{\pi}{2}\right)$ or $\left(-\frac{\pi}{2}, 1\right)$ or $\left(1, \frac{\pi}{2}\right)$ or $\left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$ ...

$\left(-\frac{\pi}{2}+k\pi, \frac{\pi}{2}+k\pi\right)$ for $k \neq 0$

and $\left(-\frac{\pi}{2}, 1\right)$ and $\left(1, \frac{\pi}{2}\right)$

In each of Problems 7 through 10, determine whether the given functions are linearly dependent or linearly independent. If they are linearly dependent, find a linear relation among them.

4.1. 9. $f_1(t) = 2t - 3$, $f_2(t) = t^2 + 1$, $f_3(t) = 2t^2 - t$, $f_4(t) = t^2 + t + 1$

$\{1, t, t^2\}$

$$c_1 f_1 + c_2 f_2 + c_3 f_3 + c_4 f_4 = 0$$

$?(c_1, c_2, c_3, c_4) \neq (0,0,0,0)$

$$c_1(2t-3) + c_2(t^2+1) + c_3(2t^2-t) + c_4(t^2+t+1) = 0$$

$\Rightarrow$ if all $c_i = 0$ then lin. independent.

if $\exists c_i \neq 0$ then lin. dependent. ← if there is

$$t^2(c_2 + 2c_3 + c_4) + t(2c_1 - c_3 + c_4) + 1(-3c_4 + c_2 + c_4) = 0t^2 + 0t + 0$$

$=0$ $\quad$ $=0$ $\quad$ $=0$

4 unknowns c_1, c_2, c_3, c_4 & 3 constraints $\rightarrow$ degree of freedom ≥ 1

$c_1 = -\frac{2}{3}$, $c_2 = -\frac{13}{3}$, $c_3 = 1$, $c_4 = \frac{7}{3}$ Solves it.

In each of Problems 21 through 24, use Abel's formula (Problem 20) to find the Wronskian of a fundamental set of solutions of the given differential equation.

4.1. 23. $y''' + \frac{2}{t}y'' - \frac{1}{t}y' + \frac{1}{t}y = 0$

Problem 20 (d) Generalize this argument to the nth order equation

$$y^{(n)} + p_1(t)y^{(n-1)} + \cdots + p_n(t)y = 0$$

with solutions $y_1, \ldots, y_n$. That is, establish Abel's formula

$$W(y_1, \ldots, y_n)(t) = c\exp\left[-\int p_1(t)\,dt\right]$$

for this case.

$$W(y_1, y_2, y_3)(t) = ce^{-\int \frac{2}{t}\,dt} = ce^{-2\ln|t| + c}$$

$$= ct^{-2}$$

In each of Problems 27 and 28, use the method of reduction of order (Problem 26) to solve the given differential equation.

4.1. 27. $(2-t)y'' + (2t-3)y' - ty + y = 0, \quad t < 2; \quad y_1(t) = e^t$

$$y = v y_1 = v e^t$$
$$y' = v' e^t + v e^t$$
$$y'' = v'' e^t + 2v' e^t + v e^t$$
$$y''' = v''' e^t + 3v'' e^t + 3v' e^t + v e^t$$

$$(2-t)(v''' e^t + 3v'' e^t + 3v' e^t + v e^t) + (2t-3)(v'' e^t + 2v' e^t + v e^t) - t(v' e^t + v e^t) + v e^t = 0$$

$$(2-t)(v''' + 3v'' + 3v' + v) + (2t-3)(v'' + 2v' + v) - t(v' + v) + v = 0$$

$$(2-t)v''' + (3-t)v'' + (6 - 3t + 4t - 6 - t)v' + (2 - t + 12t - 3 - t + 1)v$$

$$(2-t)v''' + (3-t)v'' = 0 \longrightarrow (2-t)u' + (3-t)u = 0 \quad u = v''$$

$$(2-t)v''' = (t-3)v''$$

$v'' = u \qquad \dfrac{u'}{u} = \dfrac{v'''}{v''} = \dfrac{t-3}{2-t} = -1 - \dfrac{1}{2-t}$

$\ln|2-t|$, ma $t<2$

$$\ln u = -t + \ln(2-t) + c$$

$$\int v'' = \int dt \qquad u = e^{-t} \cdot \underbrace{e^{\ln(2-t)}}_{2-t} \cdot \underbrace{e^{c}}_{c} = c\, e^{-t}(2-t)$$

$\int dt$

$$\int \underset{u}{t}\,\underset{dv}{e^{-t}}\,dt = -te^{-t} + \int e^{-t}\,dt \qquad uv - \int u\,dv$$

$dv = dt$

$u = -e^{-t}$

$$= -te^{-t} - e^{-t} + c$$

$$v' = \int c_1 e^{-t}(2-t)\,dt$$

$$= c_1 \int 2e^{-t} - te^{-t}\,dt$$

$$= -2c_1 e^{-t} - c_1\left(-te^{-t} - e^{-t} + c_2\right)$$

$$= -c_1 e^{-t} + c_1 te^{-t} - c_1 c_2$$

$$v = \int -c_1 e^{-t} + c_1 te^{-t} - c_1 c_2\,dt$$

$$= c_1 e^{-t} + c_1\left(-te^{-t} - e^{-t} + c_3\right) - c_1 c_2 t$$

$$= -c_1 te^{-t} + c_4 - c_4^* t$$

$$y = v y_1 = \left(-c_1 te^{-t} + c_{4_2} - c_3 t\right)e^t$$

$$= -c_1 t + c_2 e^t - c_3 te^t$$

$\downarrow y_2 \qquad \underbrace{\quad}_{y_1} \qquad \downarrow y_3$